Introduction to Riordan poset and its applications

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   - What is Combinatorics
   - What are combinatorialists interested in?
2. Counting and Riordan Matrix
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3. Riordan Poset
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Current Section

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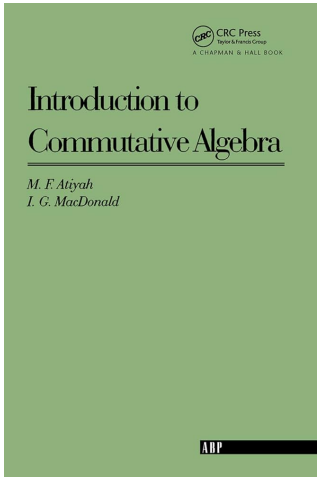
What is Combinatorics



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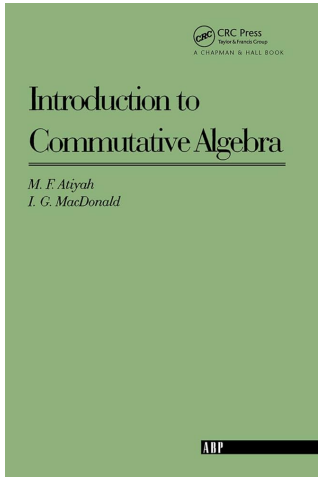
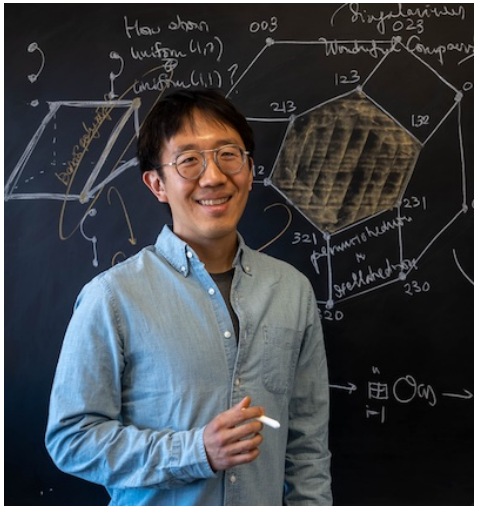


"You start out in life doing mathematics, you end up doing combinatorics."

— Ian G. Macdonald

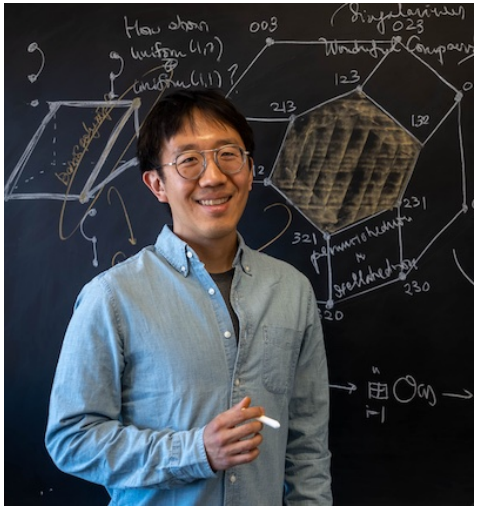
What is Combinatorics



"Combinatorics is the study of finite relations from discrete objects"

— June Huh

What is Combinatorics



"Combinatorics is the study of finite relations from discrete objects"

— June Huh

"Combinatorics is the study of the most basic and most obvious things"

— I think

What is Combinatorics

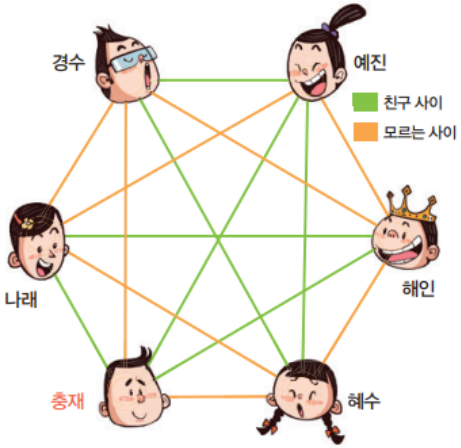
- $3 < 4$

What is Combinatorics

- $3 < 4 \rightarrow$ Pigeonhole Principle

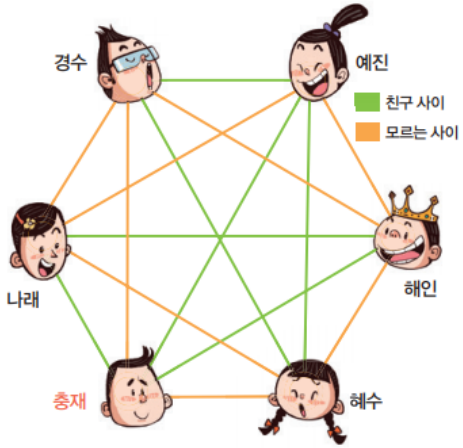
What is Combinatorics

- $3 < 4 \rightarrow$ Pigeonhole Principle $\rightarrow$ Ramsey's Theory



What is Combinatorics

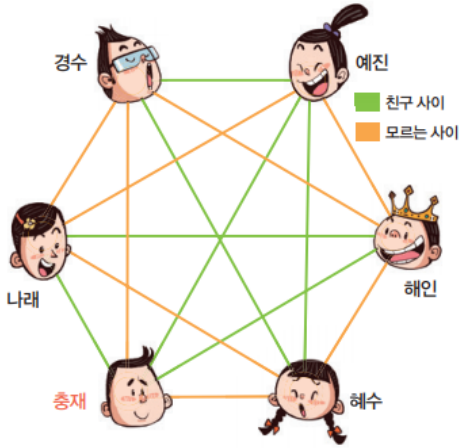
- $3 < 4 \rightarrow$ Pigeonhole Principle $\rightarrow$ Ramsey's Theory



- $|A \cup B|$

What is Combinatorics

- $3 < 4 \rightarrow$ Pigeonhole Principle $\rightarrow$ Ramsey's Theory



- $|A \cup B| \rightarrow$ Inclusion-Exclusion Principle

What is Combinatorics

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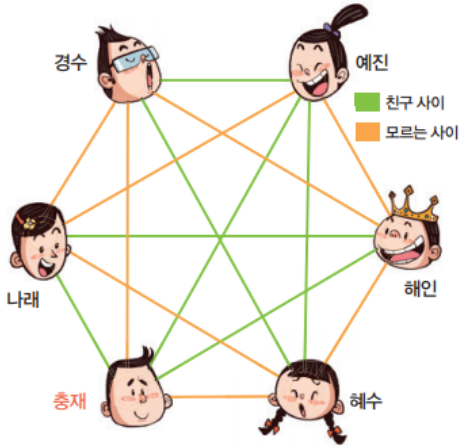
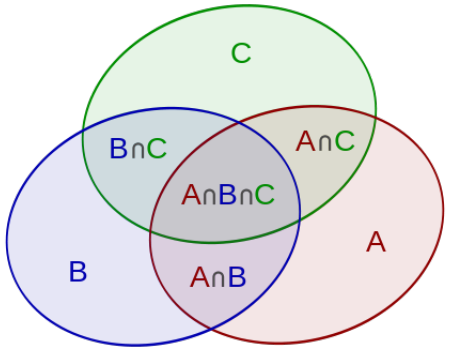


- $|A \cup B| \rightarrow$ Inclusion-Exclusion Principle $\rightarrow$ Möbius Inversion



What are combinatorialists interested in?

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(1) Counting

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(2) Subsets

$$\{\varnothing, \{x\}, \{y\}, \{z\}, \{x, y\}, \{y, z\}, \{z, x\}, X\}$$

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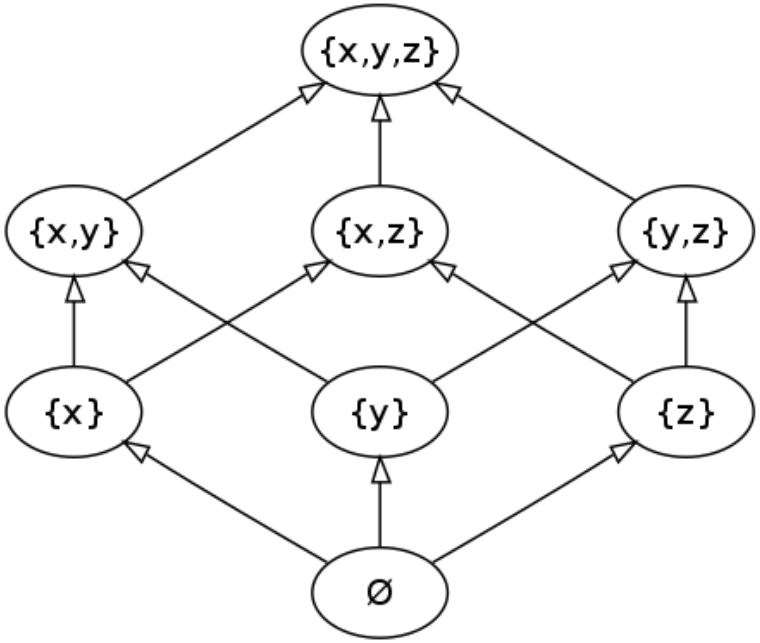

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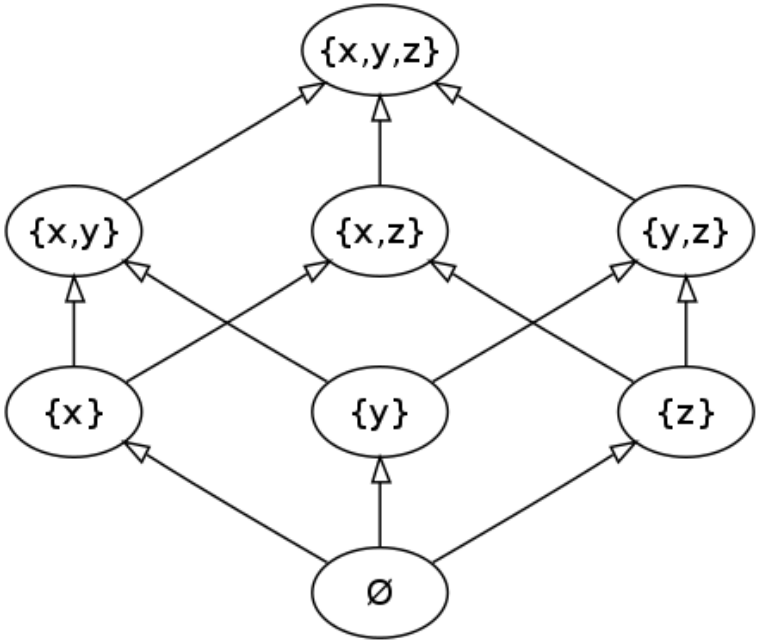

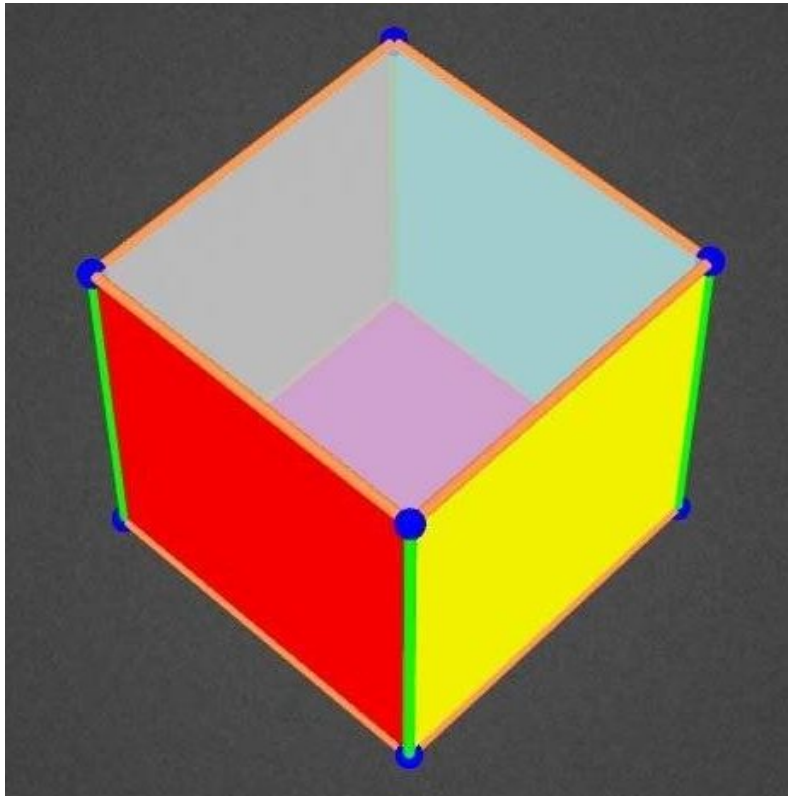

What are combinatorialists interested in?

Counting	Partially Ordered Set

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Counting

Counting is important

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- Classification

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- Classification
- $X \leftrightarrow Y$

Counting

How many objects satisfy a certain condition at **n** steps?

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In other words, what is the explicit form of a sequence a_n?

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In other words, what is the explicit form of a sequence a_n?

- Arithmetic sequence

$$a_n = a + d(n-1)$$

- Geometric sequence

$$a_n = ar^{n-1}$$

Counting

However, finding the explicit form of a sequence is very difficult

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- Recurrence relation
  ex) Fibonacci sequence: $f_n = f_{n-1} + f_{n-2}$ where $f_0, f_1 = 1$

Counting

However, finding the explicit form of a sequence is very difficult

- Recurrence relation
  ex) Fibonacci sequence: $f_n = f_{n-1} + f_{n-2}$ where $f_0, f_1 = 1$
- Generating function
  A function $f(x)$ is the (ordinary) generating function of a_n if

$$f(x) = a_0 + a_1 x + a_2 x^2 + \cdots = \sum_{i=0}^{\infty} a_i x^i.$$

ex) $a_n = 1, 1, 1, 1, \ldots$. The ordinary generating function of a_n is

$$\frac{1}{1-x} = 1 + x + x^2 + \cdots .$$

Counting

If a_n is difficult to find directly, we often use the case-by-case method.

ex) Find the number of cases in which 10 people are selected from a pool of n, and 5 or fewer men are selected.

k: the number of selected men.

$$a_n = \sum_{k=0}^{5} a_{n,k}.$$

Counting

We can draw it as a **table**

$a_{n,k}$	k				
n	0	1	2	3	a_n
0	$a_{0,0}$	$a_{0,1}$	$a_{0,2}$	$a_{0,3}$	a_0
1	$a_{1,0}$	$a_{1,1}$	$a_{1,2}$	$a_{1,3}$	a_1
2	$a_{2,0}$	$a_{2,1}$	$a_{2,2}$	$a_{2,3}$	a_2
3	$a_{3,0}$	$a_{3,1}$	$a_{3,2}$	$a_{3,3}$	a_3

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3	$a_{3,0}$	$a_{3,1}$	$a_{3,2}$	$a_{3,3}$	a_3

We can consider it as a matrix $\begin{bmatrix} a_{0,0} & a_{0,1} & a_{0,2} & a_{0,3} \\ a_{1,0} & a_{1,1} & a_{1,2} & a_{1,3} \\ a_{2,0} & a_{2,1} & a_{2,2} & a_{2,3} \\ a_{3,0} & a_{3,1} & a_{3,2} & a_{3,3} \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \end{bmatrix}$

Counting

$$\begin{bmatrix} a_{0,0} & a_{0,1} & a_{0,2} & a_{0,3} \\ a_{1,0} & a_{1,1} & a_{1,2} & a_{1,3} \\ a_{2,0} & a_{2,1} & a_{2,2} & a_{2,3} \\ a_{3,0} & a_{3,1} & a_{3,2} & a_{3,3} \end{bmatrix}$$

$$\begin{array}{cccc} \downarrow & \downarrow & \downarrow & \downarrow \\ f_0 & f_1 & f_2 & f_3 \end{array}$$

We can think of a generating function for each column.

Riordan Matrix

Definition

A *Riordan matrix* is an infinite lower triaghular matrix $A = [a_{i,j}]$ constructed from a pair of formal power series $g = 1 + g_1 t + g_2 t^2 + \cdots$ and $f = t + f_2 t + f_3 t^3 + \cdots$ by letting

$$a_{i,j} = [t^i] g f^j$$

where $[t^i]$ denotes the coefficient extraction operator.

$$A = \begin{bmatrix} | & | & | & \\ g & gf & gf^2 & \cdots \\ | & | & | & \end{bmatrix}$$

As usual, we denote the Riordan matrix by $A = (g, f)$.

Riordan Matrix

Example

The Pascal triangle is the Riordan matrix $\left(\frac{1}{1-t}, \frac{t}{1-t}\right)$.

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 \\ 1 & 3 & 3 & 1 \end{bmatrix} = \begin{bmatrix} \binom{0}{0} & \binom{0}{1} & \binom{0}{2} & \binom{0}{3} \\ \binom{1}{0} & \binom{1}{1} & \binom{1}{2} & \binom{1}{3} \\ \binom{2}{0} & \binom{2}{1} & \binom{2}{2} & \binom{2}{3} \\ \binom{3}{0} & \binom{3}{1} & \binom{3}{2} & \binom{3}{3} \end{bmatrix}$$

Riordan Matrix

$\frac{1}{1-t} = 1 + t + t^2 + \cdots$ and $\frac{t}{1-t} = t + t^2 + t^3 \cdots$.

Then,

$$\begin{aligned}
\frac{1}{1-t}\frac{t}{1-t} &= \binom{1}{1}t + \binom{2}{1}t^2 + \binom{3}{1}t^3 + \cdots \\
&= t + 2t^2 + 3t^3 + \cdots \\
\frac{1}{1-t}\left(\frac{t}{1-t}\right)^2 &= \binom{2}{2}t^2 + \binom{3}{2}t^3 + \binom{4}{2}t^4 + \cdots \\
&= t^2 + 3t^3 + 6t^4 + \cdots
\end{aligned}$$

Riordan Matrix

Definition

A sequence $A = (a_0, a_1, \ldots)$ is the *A-sequence* of the Riordan matrix $D = (g, f) = [d_{n,k}]$ if

$$d_{n+1,k+1} = a_0 d_{n,k} + a_1 d_{n,k+1} + a_2 d_{n,k+2} + \cdots = \sum_{j=0}^{\infty} a_i d_{n,k+j},$$

for all $n, k \in \mathbb{N}$.

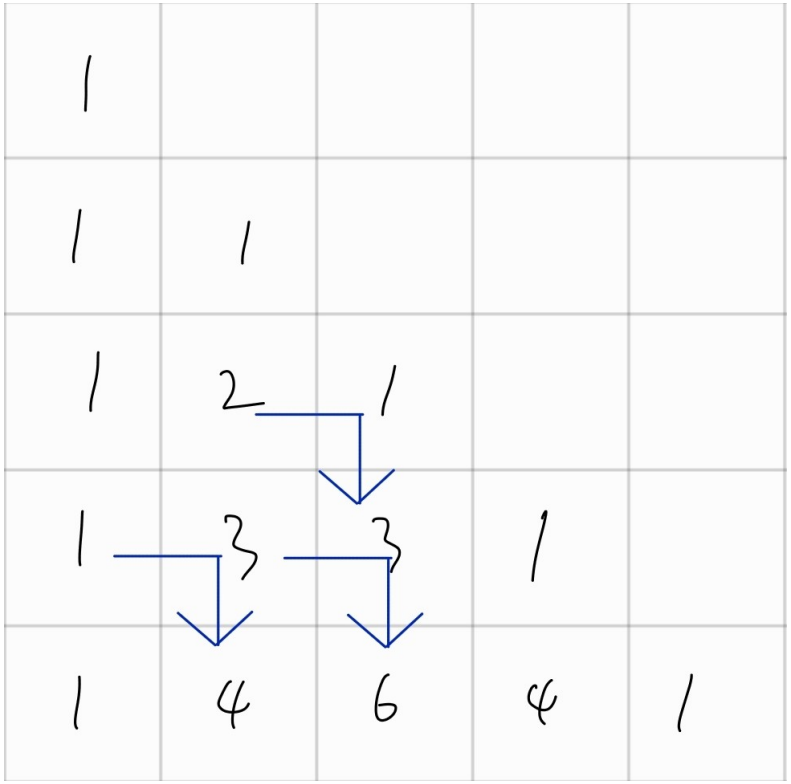
$A =$	$($	2	$,\ 3$	$,\ 1$	$,\ 0$	$,\ \cdots$	$)$
		$\times$	$\times$	$\times$	$\times$	$\times$	
		$d_{n,k}$	$d_{n,k+1}$	$d_{n,k+2}$	$d_{n,k+3}$	$\cdots$	
			$d_{n+1,k+1}$				

Riordan Matrix

Example

The A-sequence of the Pascal matrix is

$$A = (1, 1, 0, 0, 0, \ldots).$$



Riordan Matrix

Theorem

An infinite lower triangular matrix $D = (g(t), f(t)))$ is a proper Riordan matrix if and only if a sequence $A = (a_0 \neq 0, a_1, a_2, \ldots)$ is the A-sequence of D. In particular, $f(t) = tA(f(t))$ where $A(t)$ is the ordinary generating function of A-sequence A.

Recall: The Pascal matrix is a Riordan matrix $\left(\frac{1}{1-t}, \frac{t}{1-t}\right)$.

Let's obtain $\frac{t}{1-t}$ from A-sequence.
The A-sequence of the Pascal matrix is $A = (1, 1, 0, 0, 0, \ldots)$.
The oridinary generating function $A(t)$ of $(1, 1, 0, 0, 0, \ldots)$ is just $1 + t$.
Substituting this, then $f(t) = tA(f(t)) = t(1 + f(t))$.
So, $f(t) = \frac{t}{1-t}$.

Riordan Matrix

$$(g, f) \Longleftrightarrow (g, A)$$

Riordan Matrix

Fundamental Theorem of Riordan Arrays

Let $B = (b_0, b_1, \ldots)^T$ and $C = (c_0, c_1, \ldots)^T$ be column vectors and let $B(t)$ and $C(t)$ be the corresponding generating functions. If $(g, f)B = C$ then $C(t) = g(t)B(f(t))$.

Riordan Matrix

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Corollary

Let (g, f) and (d, h) be Riordan matrices. Then,

$$(g, f)(d, h) = (gd(f), h(f)).$$

Riordan Matrix

Fundamental Theorem of Riordan Arrays

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Corollary

Let (g, f) and (d, h) be Riordan matrices. Then,

$$(g, f)(d, h) = (gd(f), h(f)).$$

Theorem

Let $\mathcal{R}$ be the set of all proper Riordan matrices. Then $\mathcal{R}$ forms a **group** under the binary operation which is introduced in above corollary.

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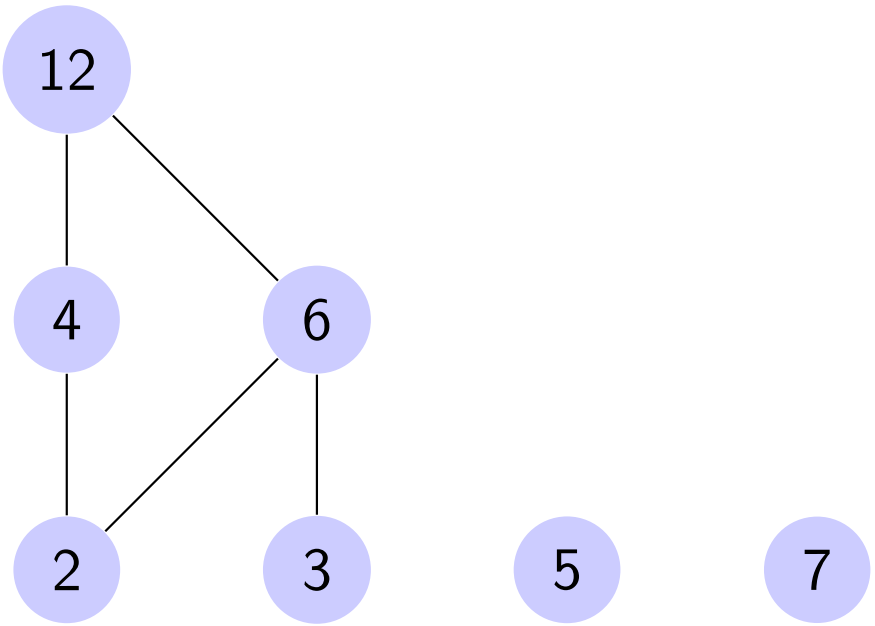
Poset

Definition

A pair $(P, \leq)$ is a partially ordered set if it satisfies

- reflexive: $x \leq x$, for all $x \in P$.
- antisymmetry: if $x \leq y$ and $y \leq x$ then $x = y$, for all $x, y \in P$.
- transitive: if $x \leq y$ and $y \leq z$, then $x \leq z$, for all $x, y, z \in P$.

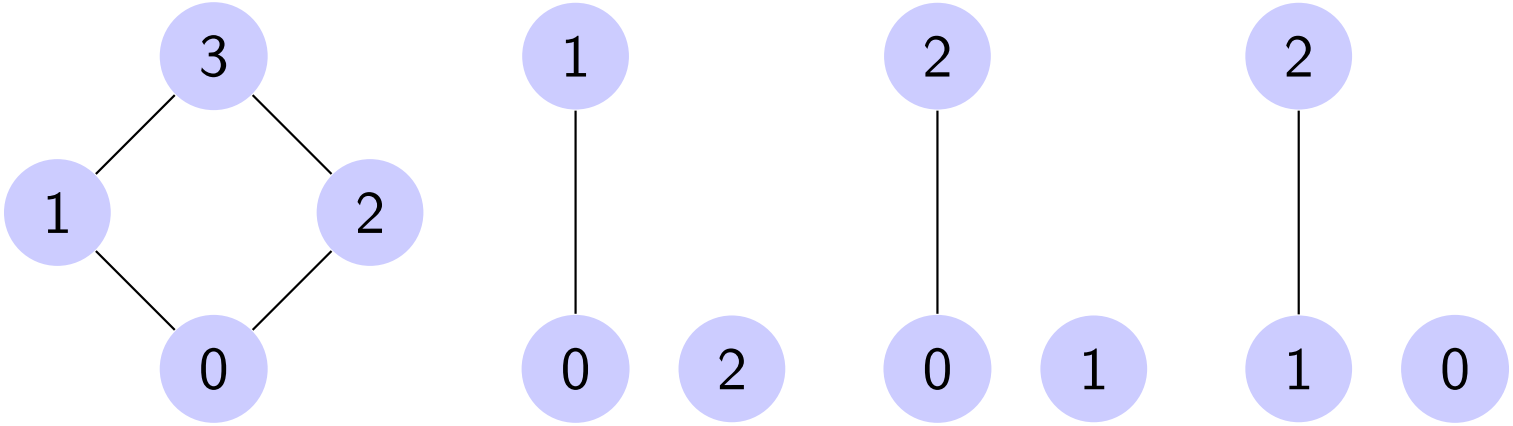
$(X, \leq) = (\{2, 3, 4, 5, 6, 7, 12\}, |)$ where $|$: divisible



Poset

Definition

Let $(P, \leq)$ be a poset where $P = \{0, 1, 2, \ldots, n-1\}$ and let $M = [a_{i,j}]$ be a $n \times n$ matrix. M is the poset matrix of P if $a_{i,j} = 1$ when $j \leq i$ in $(P, \leq)$ and $a_{i,j} = 0$ otherwise.



$$\Updownarrow \quad \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix} \qquad \Updownarrow \quad \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad \Updownarrow \quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \qquad \Updownarrow \quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

Riordan Poset

Definition

An infinite $(0,1)-$matrix A is a *Riordan poset matrix* if A is a Riordan matrix over $\mathbb{Z}_2$ and poset matrix.

Riordan Poset

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An infinite $(0,1)-$matrix A is a *Riordan poset matrix* if A is a Riordan matrix over $\mathbb{Z}_2$ and poset matrix.

Theorem

The *binary Pascal matrix* P is a matrix whose (i,j)-entry is $\binom{i}{j}$ mod 2.

$$P = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & \cdots \\ 1 & 1 & 0 & 0 & 0 & \cdots \\ 1 & 2 & 1 & 0 & 0 & \cdots \\ 1 & 3 & 3 & 1 & 0 & \cdots \\ 1 & 4 & 6 & 4 & 1 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix} \pmod 2 = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & \cdots \\ 1 & 1 & 0 & 0 & 0 & \cdots \\ 1 & 0 & 1 & 0 & 0 & \cdots \\ 1 & 1 & 1 & 1 & 0 & \cdots \\ 1 & 0 & 0 & 0 & 1 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix}$$

Then, P is a Riordan poset matrix.

Riordan Poset

Definition

A Toeplitz poset matrix is a poset matrix in which each descending diagonal from left to right is constant.

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ g_1 & 1 & 0 & 0 \\ g_2 & g_1 & 1 & 0 \\ g_3 & g_2 & g_1 & 1 \end{bmatrix}$$

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Every Toeplitz poset matrix is a Riordan poset matrix of form $(g(t), t)$.

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Theorem

Let $g = 1 + g_1 t + g_2 t^2 + \cdots \in \mathbb{Z}_2[[t]]$. Let (g, t) is a Riordan poset matrix if and only if $g_{a+b} = 1$ whenever $g_a = 1$ and $g_b = 1$ for any integers $a \geq b \geq 1$.

Riordan Poset

Recall:

- The A-sequence of the Pascal matrix is $(1, 1, 0, 0, 0, \ldots)$.

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  We call Riordan poset matrices which has the A-sequence $(1, 1, 0, 0, 0, \ldots)$ *Pascal type*

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  We call Riordan poset matrices which has the A-sequence $(1, 0, 0, 0, 0, \ldots)$ *Toeplitz type*

Conjecture

The A-sequence of an infinite Riordan poset matrix is either $(1, 0, 0, ...)$ or $(1, 1, 0, 0, ...)$.

Thank you